

Consultancy Services for Design and Supervision for STP, Sewer & TE Networks Systems in Ibri

Client: Nama Water Services (NWS / OWWSC)

CONCEPT SCREENING REPORT

Pre-Kickoff GIS Assessment — Working Set W1, Revision R0

Renardet S.A. & Partners | Project 2621
20 July 2026

Screening-grade outputs based on uncontrolled data (TOR §H.14). Not for design. Report styling provisional pending sample report (GAP-4).

Table of Contents

Table of Contents	2
1. Introduction	3
1.1 Design basis documents	3
2. Scope Highlights Binding This Work	3
3. Design Criteria Applied (Standards Extract).....	3
3.1 Gravity network hydraulics (PAM-GUD-203).....	3
3.2 Flow estimation parameters (PAM-GUD-201).....	3
4. Study Area and Data	4
5. Topography and Gravity Feasibility	5
6. Road Network Analysis	6
7. Concept Trunk and Sewer Zones	7
8. Flow Estimation (Ultimate / Saturated).....	8
9. Lifting Stations vs New/Satellite STP	9
9.1 Screening method and results	9
9.2 Criteria — when does a pocket justify an SLS, and when a satellite STP?.....	9
10. Wadi Crossings.....	9
11. Open Gaps and Kickoff Questions	10
12. Next Steps (W2)	10

1. Introduction

Nama Water Services (NWS/OWWSC) intends to design and construct wastewater, treated-effluent (TE) networks and STP capacity for Ibri Wilayat (Tender T/2719110/2025). The objective is to verify the RG Masterplan, determine the ultimate sewage flow for a design horizon of project completion + 25 years (or saturation), and deliver concept, preliminary and detailed designs and tenders (TOR §3).

This report is a pre-kickoff concept screening prepared from the client-provided information folder. Per TOR §H.14 such data is indicative only; every result herein is screening-grade and will be superseded by topographic survey, data collection and hydraulic modelling (SewerGEMS) during the concept stage. The terrain model used is a coarse DSM (surface model, includes buildings/vegetation); design-grade conclusions are expressly excluded.

1.1 Design basis documents

- PAM-GUD-203 Wastewater Design Guidelines v1.0 (Rev 01) — primary standard, cited as p##.
- PAM-GUD-201 General Design Guidelines v1.0 (Rev 01) — planning parameters, cited as G1-p##.
- TOR Section 03 (Scope of Work), Tender T/2719110/2025.

2. Scope Highlights Binding This Work

Item	Requirement	Ref
Gravity preference	Gravity connection preferred; pumping only where gravity not realistic or too costly; both options to be considered	TOR p10
Topography	Layout shall avoid pumping and utilise gravity as much as practicable using contour mapping	TOR p12
Coverage	All plots within boundary (built, open, under construction) to be serviced	TOR §H.2
Options	Minimum three options for WW network, TE network and each STP	TOR p13, §H.18
STP threshold	STP preliminary design + EPC tender in consultant scope only if Phase I < 20,000 m ³ /d	TOR p3, p5
Model years	Start (TBA), 2030, 2055 and ultimate; SewerGEMS/WaterGEMS deliverables	TOR p14
Inverted siphons	To be avoided; last resort only	TOR p12

3. Design Criteria Applied (Standards Extract)

3.1 Gravity network hydraulics (PAM-GUD-203)

Criterion	Value	Ref
Design formula	Colebrook-White ($k_s = 1.5 \text{ mm}$) or Manning; approved software	p24
Min self-cleansing velocity	0.75 m/s at peak flow (0.90 preferred); tractive-force check at heads	p26-27
Max velocity	3.0 m/s	p27
Depth of flow at peak	$d/D \leq 0.65$ ($D \leq 350 \text{ mm}$); $d/D \leq 0.50$ ($D > 350 \text{ mm}$)	p27
Min gradient (Table 11)	DN200: 5.0 DN315: 2.7 DN500: 1.55 DN700: 1.0 $DN \geq 900$: 0.75 mm/m	p29
Cover	Min 1.3 m to crown (0.5 m w/ concrete protection); max ~10-12 m, beyond -> pumping	p33
Manhole spacing	100 m (DN200-315) to 200 m ($> DN1400$)	p30
Trunk main definition	$D > 800 \text{ mm}$, $> 1,000 \text{ m}$ without connections, upstream of STP/main PS	p35
Wadis	Pipelines and chambers in wadis / washout areas to be avoided	p30, p33

3.2 Flow estimation parameters (PAM-GUD-201)

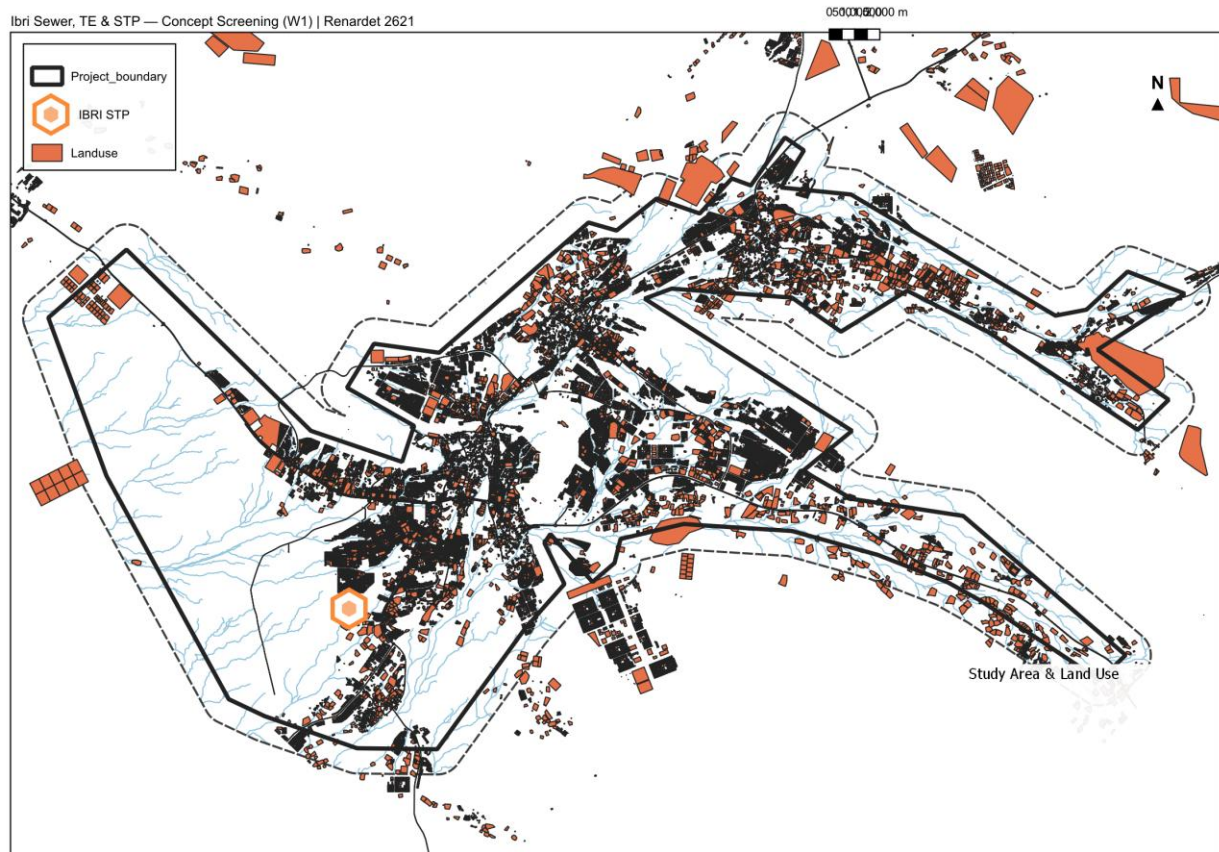
Parameter	Value	Ref
Population from plots	plots x properties/plot x occupancy rate (NCSI)	G1-p58
Domestic consumption, Adh Dhahirah	164 l/c/d (IMP 2024 baseline)	G1-p60
Non-domestic / governmental	+22% / +14% of domestic LPCD	G1-p60
Return to sewer	85% domestic & tanker; 54% non-domestic/governmental	G1-p71
Peak factor	Peltier PfWW = $1.5 + 1/\sqrt{Qm[l/s]}$ (cap 5.0); Merrimack alternative	G1-p71-72
Infiltration (new networks)	720 L/d per km of sewer	G1-p72
STP design margin	+10% operational safety allowance	G1-p73
STP organic load	>= 60 g BOD5 and 80 g TSS /cap/day	GUD-203 p74

4. Study Area and Data

Project boundary: 439.8 km², single polygon (EPSG:32640). Within it, serviceable land use concentrates in one principal agglomeration around Ibri town plus satellite settlements. The existing Ibri STP lies at the south-west edge of town at ~327.6 m.

Land use (in boundary)	Plots	Area (km ²)
Residential	43,722	32.2
Agriculture	4,310	36.8
Governmental	2,976	16.3
Commercial	3,057	2.4
Industry	466	0.9
Not classified	3,991	7.5

Data quality flags: NSA DEM is a coarse 4 m DSM (survey ongoing) — screening only. Road centreline dataset is regional (57,584 segments) with no functional-class attributes; dual carriageways are represented as two parallel polylines. Existing sewer as-built extents (Figure F2) are pending georeferencing.



Map M1 — Study area, land use and existing STP (layout per project template)

5. Topography and Gravity Feasibility

Ground elevations at 43,297 serviceable plots of the main agglomeration were sampled against the existing STP inlet area (327.6 m). The main town (median 374.3 m) commands a median straight-line available grade of 3.6 m/km towards the STP — several times the DN $\geq$ 900 trunk minimum of 0.75 mm/m (p29). The main-trunk long section is monotonic with only 2-4 m local dips.

Indicator	Value
STP ground level	327.6 m
Main cluster median elevation	374.3 m
Median available grade to STP	3.6 m/km (p25: 3.2)
Plots with < 1.0 m/km available	2.8%
Plots below STP + 5 m	5.4%

Conclusion: Ibri is a gravity town. The concept should target gravity conveyance to the existing STP location for ~95% of demand, with lifting confined to identified low or distant pockets (Section 8). This aligns with the TOR gravity preference and GUD-203 topography-driven siting.

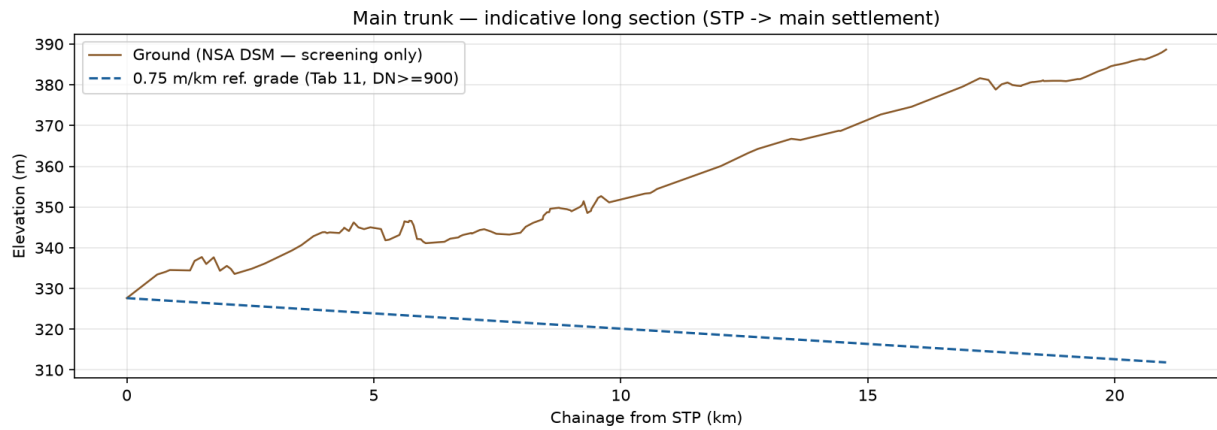
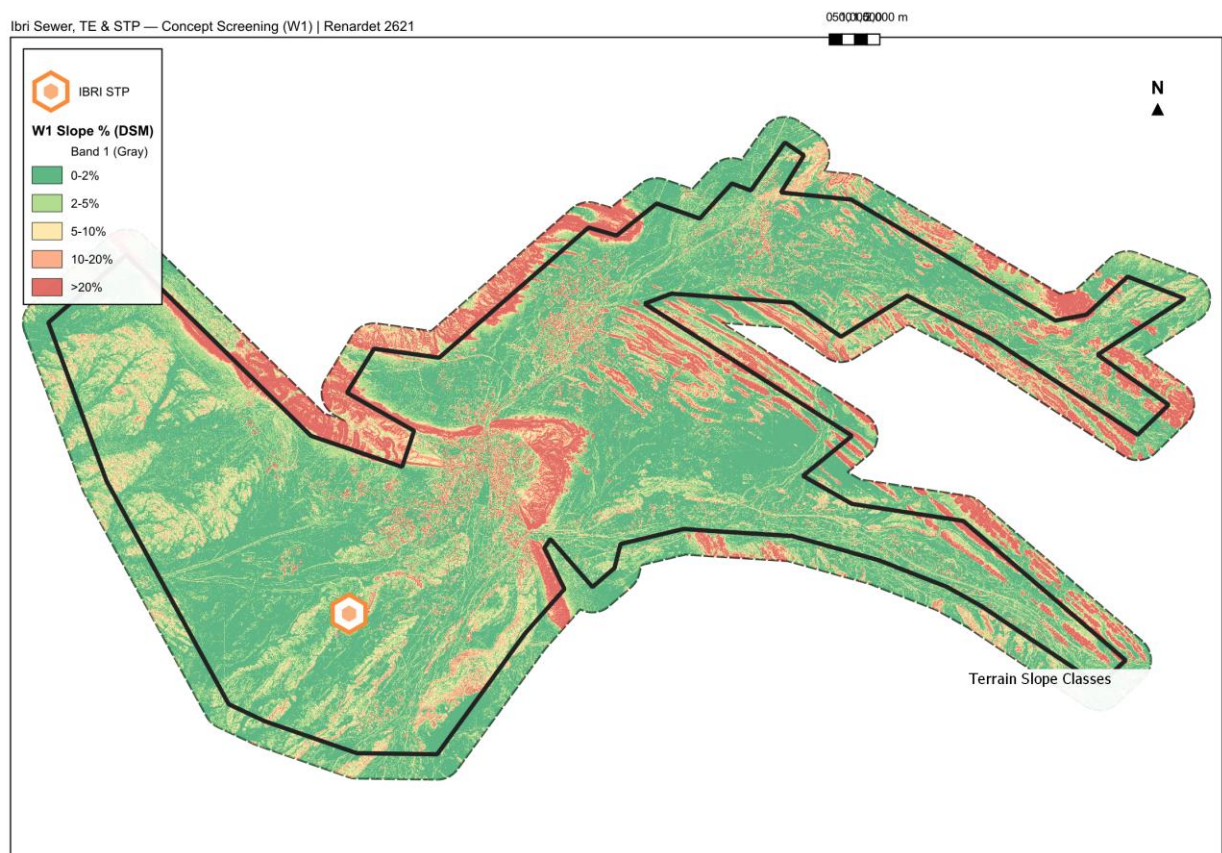


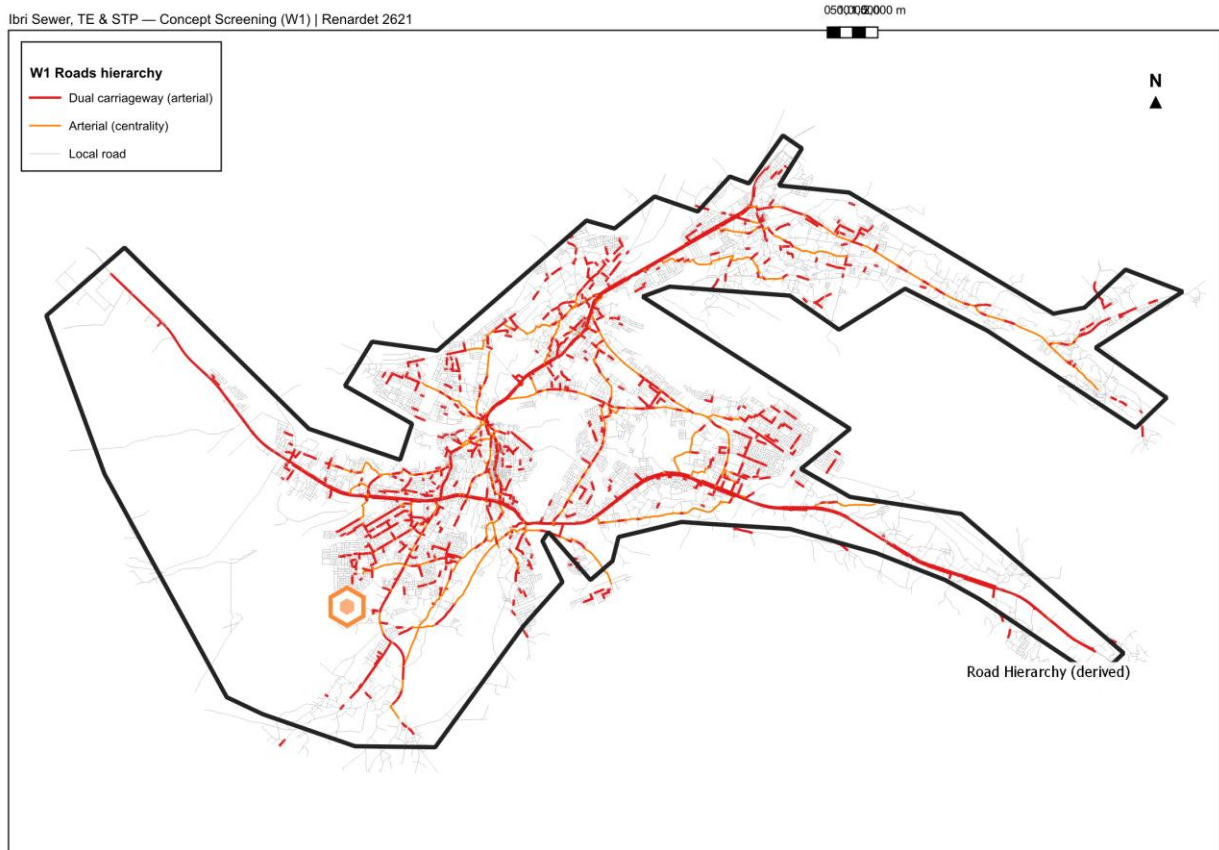
Figure — Main trunk indicative long section (DSM ground; screening)



Map M2 — Terrain slope classes (DSM)

6. Road Network Analysis

The road dataset holds no hierarchy attributes, so arterials were derived: (i) edge betweenness centrality on the routed graph (top decile), and (ii) geometric detection of dual-carriageway pairs (parallel polylines 6-45 m apart, bearing within 12 deg) gated by centrality to reject residential grid false positives. Result: 3,375 arterial edges of 17,862, forming a connected skeleton of the dual-carriageway corridors and central links.

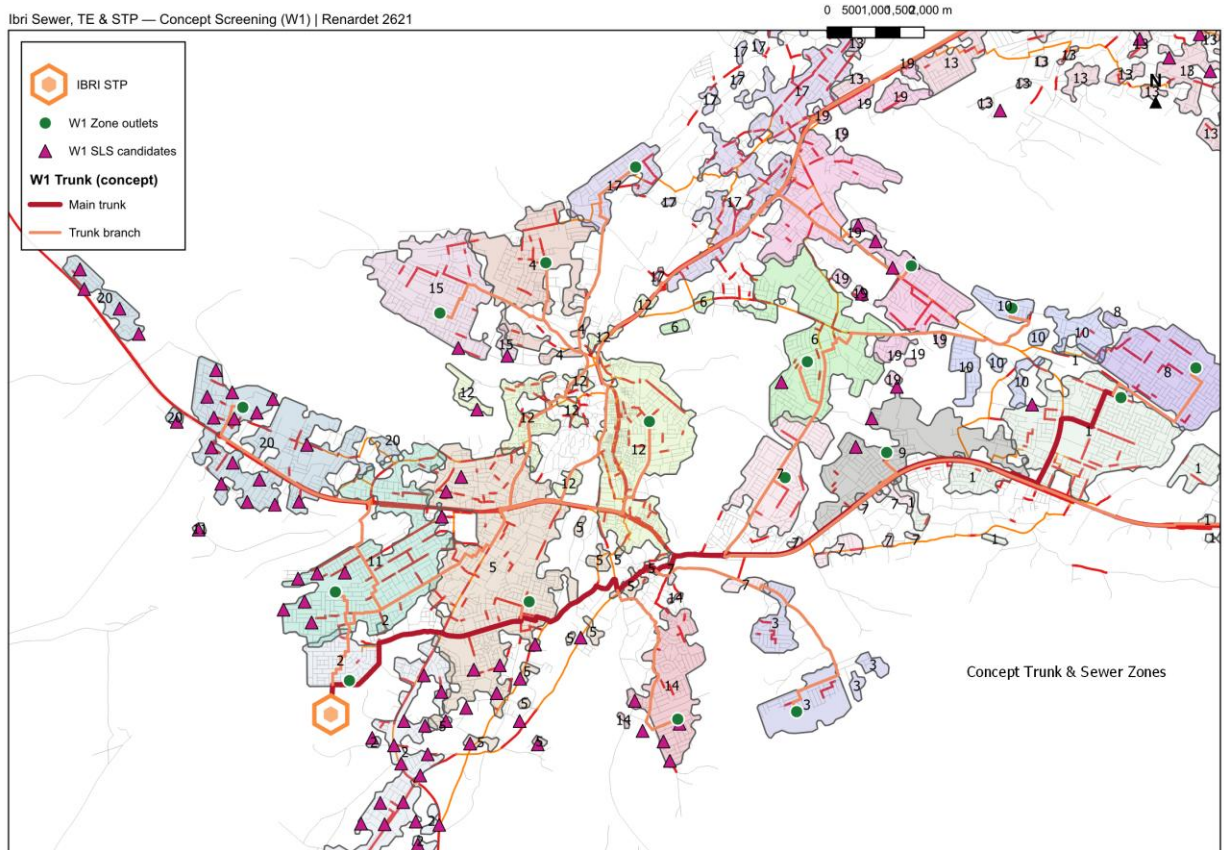


Map M3 — Derived road hierarchy

7. Concept Trunk and Sewer Zones

Trunk routing used least-cost paths on the road graph with arterial discounting (dual 0.55, arterial 0.70, local 1.00), which keeps the trunk on near-straight main-road corridors per the routing preference, dropping to internal roads only where connectivity requires. The trunk tree connects 20 zone outlets to the STP: main trunk 21.0 km, branches 114.3 km (135.3 km total).

Zones are road-network territories: every network node is assigned to its nearest outlet by in-network distance, weighted by plot density — not by DEM watershed — so zone boundaries follow the road fabric and plot clusters as instructed, with terrain applied afterwards as the SLS screening constraint. Each zone drains to exactly one outlet on the trunk.



Map M4 — Concept trunk, zones, outlets and SLS candidates

8. Flow Estimation (Ultimate / Saturated)

Flows follow the GUD-201 chain of Section 3.2 applied to all serviceable plots (TOR requires all plots serviced). Occupancy is a tagged assumption of 6.0 persons/housing unit and 1 property/plot pending NCSI figures [GAP-5]; totals scale linearly with it.

Zone	Plots	Pop*	Qadf (m3/d)	PF	Qpeak (m3/d)	Sewer (km)
1	4294	23346	4132.1	1.64	6795.7	185.25
2	3900	19392	3437.1	1.66	5700.6	160.6
3	1573	8220	1428.2	1.75	2493.6	28.12
4	1880	10008	1756.0	1.72	3023.4	58.02
5	3973	21336	3796.4	1.65	6267.3	197.12
6	2562	11232	1977.5	1.71	3379.6	74.56
7	1864	9624	1718.5	1.72	2963.1	97.34
8	2530	14106	2453.5	1.69	4140.7	51.96
9	1851	9528	1667.2	1.73	2880.3	48.88
10	1391	7740	1350.5	1.75	2367.3	34.42
11	2777	14832	2616.9	1.68	4400.8	106.18
12	2929	14916	2694.8	1.68	4524.8	194.42
13	3538	18816	3400.7	1.66	5643.0	246.98
14	1302	6660	1179.1	1.77	2087.9	53.33
15	1878	9870	1732.5	1.72	2985.6	58.25
16	4634	24762	4418.2	1.64	7245.1	245.73

Zone	Plots	Pop*	Qadf (m3/d)	PF	Qpeak (m3/d)	Sewer (km)
17	2863	14928	2681.3	1.68	4503.2	172.73
18	1669	8976	1679.1	1.73	2899.5	196.72
19	3205	17436	3068.9	1.67	5118.3	114.5
20	2890	14028	2531.7	1.68	4265.2	179.09

* Population at OR = 6.0 [GAP-5]. Qadf includes 720 L/d/km infiltration; PF = Peltier.

Totals: Qadf ~ 49,700 m3/d ultimate saturated (+10% STP margin -> ~54,700 m3/d); peak ~ 83,700 m3/d. Commercial implication: ultimate capacity far exceeds the 20,000 m3/d threshold of TOR p3 — STP Phase I sizing and phasing become the pivotal concept decision. A phased build-out (development-percentage per GUD-201 G1-p59) for 2030/2055 model years is required before any Phase I capacity is fixed; present-day connected flow will be a fraction of saturation.

9. Lifting Stations vs New/Satellite STP

9.1 Screening method and results

For every populated node, the invert profile to the STP was accumulated along the routing tree at Table 11 minimum gradients (diameter class from cumulative upstream plots), riding at minimum cover (1.3 m + pipe) where ground falls. Nodes whose route exceeds the 12 m maximum cover (p33) cannot reach the STP by gravity: 1,087 nodes (~8% of load), clustering into 125 candidate pockets; the ten largest hold 160-270 plots each at 13-22 m required depth.

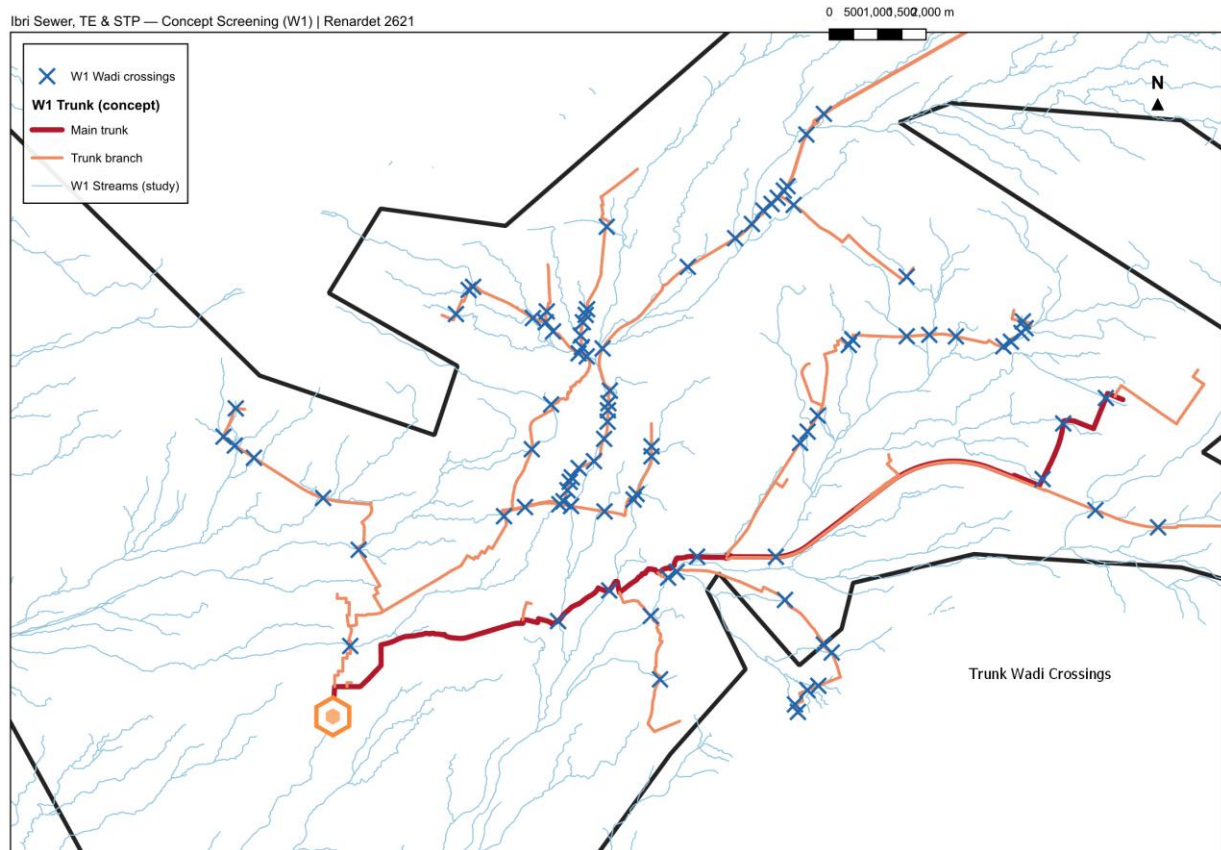
9.2 Criteria — when does a pocket justify an SLS, and when a satellite STP?

Trigger	Criterion	Source
SLS becomes necessary	Excavation cost prohibitive / cover > 10-12 m	GUD-203 p33
SLS siting	Hydraulics-driven; force-main energy-economic study; flooded suction	GUD-203 p38
Avoid SLS where possible	Topography considerations to limit pumping (STP siting criterion f)	GUD-203 p63
Satellite/new STP candidate	Remote cluster where trunk length and lift render conveyance LCC above local treatment LCC; STP >= Small category 500 m3/d viable	GUD-203 p65; TOR 3-options
Decentralised threshold (remote areas)	Compact decentralised units allowed for remote areas per GUD-201 §8	G1-p80-83
Comparison metric	Whole-life cost (CAPEX+OPEX/LCCA) across >= 3 options	G1-p95-97; TOR p13

Applied to Ibri: the eastern satellite cluster (~1,144 plots at 31.7 km network distance, +164 m above STP) is the prime satellite-STP/decentralised candidate — conveyance there is a Small-STP-scale flow over a trunk longer than the whole main town system. The 125 in-town pockets are SLS-scale questions, to be rationalised (merged/eliminated) during concept design as local sewers re-route; screening deliberately over-detects.

10. Wadi Crossings

The concept trunk crosses mapped stream lines 108 times (7 on the main trunk). Binding criteria: DI pipe across crossing +15 m each side with restrained joints (G1-p86); min cover 2.0 m in soft soil (G1-p86) / 1.5 m at force-main crossings (p52); isolation + air valves both sides, washout at low point; no chambers or markers in wadi bed or embankments (G1-p86, p30); protection per PAM-STD-404; flood data (1:20/50/100) and MoAFWR approvals (G1-p85). Stream lines derive from the DSM and are indicative; crossing inventory to be re-based on survey.



Map M5 — Trunk wadi crossings vs stream network

11. Open Gaps and Kickoff Questions

- GAP-4: sample report styling file empty — this report uses provisional styling.
- GAP-5: NCSI occupancy rate and properties/plot — flows scale with it; request via NWS.
- GAP-6: existing sewer as-built (Fig. F2) extents to be georeferenced; zones overlapping served areas to be verified.
- GAP-7: existing Ibri STP capacity, headworks invert and spare capacity; integration old-new STP intent.
- Model start year confirmation; 2030/2055/ultimate horizons; per-capita validation vs IMP (G1-p59).
- MoHUP land bank for new STP / SLS sites; F3 redesign-vs-new boundaries.

12. Next Steps (W2)

- Rationalise SLS pockets and re-route local networks; derive 3 concept options (all-gravity-max, hybrid, satellite-STP east).
- Phased flow projection 2030/2055/ultimate once NCSI/IMP figures received; SewerGEMS model seed.
- Georeference F2/F3; integrate existing network; STP phasing and land-take check (GUD-203 Tab 28 footprints).
- Restyle report per sample; populate NWS-format deliverable structure.